STUDENT MENTORSHIP & PROGRESS TRACKER

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND BUSINESS SYSTEMS



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(An Autonomous Institution, Affiliated to Anna University, Chennai)
Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

OCTOBER 2026

RAJALAKSHMI ENGINEERING COLLEGE
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BONAFIDE CERTIFICATE

Certified that this project report titled "STUDENT MENTORSHIP & PROGRESS TRACKER" is the bonafide work of SOWBARNIGA A [241401104], who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

I, SOWBARNIGA A (241401104), hereby declare that the project report titled "STUDENT MENTORSHIP & PROGRESS TRACKER", submitted for the End Semester Examination of CB23F35 – Low-Code Application Development using Zoho Creator, is a record of the project work carried out by me. The application described in this report was developed using Zoho Creator based on the implementation details and testing observations recorded during the project.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Zoho Creator documentation and other external resources used in preparing this report are acknowledged in the references.

Place: Chennai
Date: [3/11/2026]

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I also acknowledge Zoho Corporation for the Zoho Creator platform and its official documentation, which supported the implementation of forms, reports, workflows and Deluge scripting. Finally, I thank my family and friends for their encouragement.

SOWBARNIGA A

ABSTRACT

Student mentoring is a vital academic-support mechanism in higher education institutions, involving tracking student progress, assigning faculty mentors, scheduling counseling sessions, and evaluating academic performance. When these activities are tracked through separate spreadsheets, paper registers and manual records, faculty coordinators face fragmented data, repeated record entries, delayed meeting follow-ups and limited visibility into student development.

The Student Mentorship & Progress Tracker is a low-code web application developed using Zoho Creator. It contains six structured forms—Student_Registration_Form, Mentor_Registration_Form, Mentor_Allocation_Form, Academic_Progress_Form, Mentoring_Meeting_Feedback_Form and Student_CRM_Form—linked through relational lookup fields. Twelve reports including list views, meeting calendar and performance pivot charts alongside an interactive Student_Mentorship_Dashboard provide operational visibility. Four automated workflows manage registration confirmation emails, meeting alerts, automatic CRM lead creation and lead conversion updates. Deluge scripting automates data transfer across forms and sends timely notifications.

Testing recorded successful student registration, automated welcome email delivery, mentor-student lookup association, meeting scheduling on calendar, performance score tracking, and dashboard metric synchronization. The dashboard snapshot displayed 27 total records, including active students, faculty mentors, mentor allocations, and documented mentoring sessions. A built-in Zia prediction model estimates overall student performance categories based on academic criteria. Production deployment, role permissions, and formal security testing were validated within the Zoho Creator environment.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Student Mentorship; Academic Tracking

PROJECT AT A GLANCE

Attribute	Details
Project Title	Student Mentorship & Progress Tracker
Application Domain	Higher Education Operations / Student Mentorship
Student Name & Register No.	Sowbarniga A – [241401104]
Zoho Creator Edition	CB23F35 – Low-Code Application Development using Zoho Creator
Application Name (Link Name)	Student Mentorship & Progress Tracker
Published / Live URL	https://sowbarnigaa2024csbsra76.zohocreatorportal.in
Demo Credentials (per role)	Administrator, Mentor, Student
Number of Forms / Subforms	6 Forms / 0 Subforms
Number of Reports (by type)	List: 8 | Kanban: 0 | Calendar: 1 | Pivot: 3 | Total: 12
Number of Pages / Dashboards	1 Dashboard – Student_Mentorship_Dashboard
Number of Workflows	Form: 4 | Schedule: 0 | Approval: 0 | Blueprint: 0 | Batch: 0
Deluge Functions / Custom Actions	4 workflow automation scripts (welcome email, meeting notification, CRM lead sync, status update)
Integrations	Zoho Creator email notification functionality, Zia AI Prediction model

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
CRUD	Create, Read, Update and Delete
DFD	Data Flow Diagram
ER	Entity–Relationship
HR	Human Resources
KPI	Key Performance Indicator
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
UI	User Interface
URL	Uniform Resource Locator
Zoho	Zoho Corporation platform name
Deluge	Data Enriched Language for the Universal Grid Environment
API	Application Programming Interface
ID	Identifier

CHAPTER 1
INTRODUCTION

1.1 Background

Student mentoring is the structured academic counseling process through which an enrolled student is guided by an appointed faculty mentor across academic terms. The mentoring process commonly includes collecting student personal details, tracking academic progress across internal examinations, scheduling periodic review meetings, identifying learning difficulties, and recommending targeted academic improvements. Mentor coordinators and department faculty require an organised, automated record-keeping system to coordinate these activities smoothly. This project models the complete student mentoring process as a dedicated low-code application using Zoho Creator.

1.2 Problem Statement

Traditional mentoring commonly relies on separate spreadsheets, paper counseling registers, and ad-hoc email communications. These disconnected methods fragment student academic records, make it difficult to schedule meetings and follow up on struggling students, and increase administrative effort for faculty. A centralised application is required to maintain student and mentor records, link each allocation and meeting to the relevant student, monitor performance, and provide timely notifications.

- Student personal data, academic marks, and mentor allocations are maintained separately, causing repeated data entry.
- Meeting schedules, counseling notes, and progress status cannot be monitored effectively from a single interface.
- Manual tracking delays follow-up reviews and limits timely intervention for students requiring academic guidance.

1.3 Objectives

- To create structured forms for student registration, mentor registration, mentor allocation, academic performance tracking, and mentoring meeting logs.
- To automate welcome emails, meeting notifications, and CRM lead creation using Zoho Creator workflows and Deluge.
- To provide list reports, calendar views, pivot charts, and an interactive dashboard for monitoring student mentoring progress.

- To integrate Zoho Creator email functionality and Zia AI performance prediction to enhance academic tracking.
- To test the implemented forms, reports, workflows, and dashboard, ensuring reliable application deployment.

1.4 Scope of the Project

In scope: Six core forms (Student_Registration_Form, Mentor_Registration_Form, Mentor_Allocation_Form, Academic_Progress_Form, Mentoring_Meeting_Feedback_Form, Student_CRM_Form); relational lookup relationships; twelve reports including list, calendar, pivot table, and chart views; Student_Mentorship_Dashboard; automated confirmation email, meeting alert, and CRM lead sync workflows; and Zia AI student performance prediction. Out of scope or not verified: external university ERP integration, online fee payment gateways, SMS gateway integrations, and native offline mobile sync.

1.5 Justification for a Low-Code Approach

Zoho Creator was selected because it provides visual form creation, relational lookup fields, multi-view reports, customizable dashboard pages, and workflow automation within a unified low-code cloud environment. Deluge scripting adds tailored business logic and notification actions where visual configuration alone is insufficient. This allowed the project to deliver a robust, automated student mentoring solution with minimal development time compared to traditional full-stack development.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security, and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In typical academic departments, faculty mentors maintain student details, meeting minutes, and marks across separate Excel sheets or manual registers. A typical workflow involves collecting student data, assigning mentors, scheduling mentoring sessions, recording academic concerns, and manually consolidating progress reports at the end of each semester.

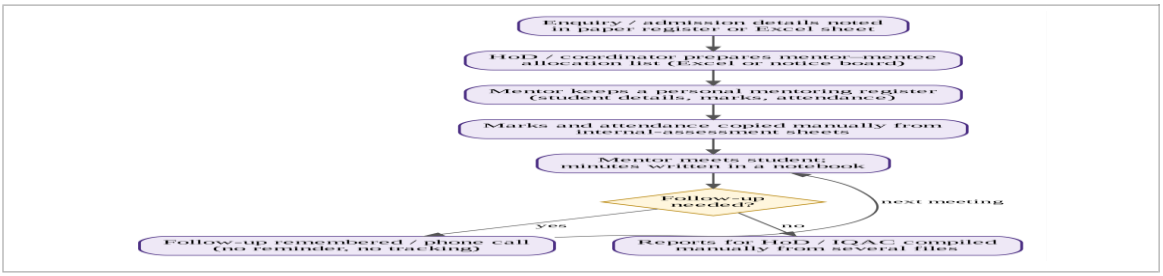


Figure 2.1: As-Is Student Mentorship Process Flow

2.2 Limitations of the Existing System

- Student personal records, academic marks, and counseling logs are fragmented across separate files.
- Meeting schedules and follow-up reminders require manual coordination and are frequently delayed.
- There is no centralized dashboard providing real-time visibility into overall student performance.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development effort	Requires separate UI, backend, database and deployment components.	Forms, reports, pages and workflows are configured in one platform.
Skills required	Programming, database, frontend and backend skills.	Visual configuration with Deluge for custom logic.

Data management	Database schema and CRUD interfaces must be built manually.	Creator forms and lookup fields provide native relational data capture.
Automation	Requires custom backend jobs or cron scheduling services.	Creator workflows automate email notifications and field updates natively.
Reporting	Reports and analytics dashboards must be developed separately.	List reports, calendar views, charts and dashboard pages are built-in.
Maintenance	Codebase, APIs and server infrastructure require ongoing upkeep.	Application configuration and Deluge scripts are maintained in the cloud.
Project fit	Flexible but involves excessive overhead for departmental workflows.	Highly suitable for an agile student mentoring and progress tracker.

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source	Title / Product	Key Findings	Relevance to this Project
1	Zoho Corporation	Understanding Forms	Creator forms capture and store application records.	Supports Student, Mentor, and Meeting forms.
2	Zoho Corporation	Understand Lookup Field	Lookup fields connect a record to another form record.	Supports student-to-mentor allocation lookups.
3	Zoho Corporation	Understanding Pages	Pages combine application components for user-facing views.	Supports the Student_Mentorship_Dashboard page.
4	Zoho Corporation	Understanding Reports in a Page	Reports can be embedded directly in Creator pages.	Supports embedding meeting and student reports.
5	Zoho Corporation	Accessing Form Fields using input	Deluge accesses and updates form fields through input keyword.	Supports automated CRM insertion and email scripts.

6	Zoho Corporation	System Variables & Workflows	System date values and automated events support business logic.	Supports registration confirmation and meeting alerts.
7	Zoho Corporation	Zia AI Modeler	Built-in machine learning models predict categorical outcomes.	Supports student academic performance prediction.

2.5 Emerging Trends in Low-Code Development

The project demonstrates key low-code trends including visual application building, workflow automation, and AI-assisted performance prediction. Zoho Creator Zia AI was utilized to model student academic performance and forecast overall grades based on historical scores and attendance. Low-code platforms dramatically reduce implementation time while Deluge scripting ensures complete programmatic control over workflow actions, automated communications, and cross-form transactions.

2.6 Proposed System

The proposed Student Mentorship & Progress Tracker centralises student profiles, mentor registrations, allocation records, academic progress scores, mentoring meeting logs, and student enquiries. Lookup fields establish relational integrity across forms. Twelve reports provide diverse data perspectives (tabular, calendar, and pivot charts). An interactive dashboard displays key metrics, distribution charts, and embedded reports. Workflows automate registration emails, meeting alerts, and CRM synchronisation.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator / Mentor Coordinator	Primary administrative user managing the overall system.	Register students and mentors, allocate mentors, oversee dashboard metrics, and configure permissions.
Faculty Mentor	Academic staff mentoring assigned students.	Enter meeting counseling notes, track student marks, record attendance, and view allocated student reports.
Student Mentee	Student enrolled in the mentorship programme.	Receive registration confirmation, view allocated mentor, and check meeting feedback and scheduled dates.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow staff to register students with personal and academic details in Student_Registration_Form.	Admin / Staff	High
FR-02	The system shall automatically send an email confirmation upon successful student registration.	System / Deluge	High
FR-03	The system shall allow staff to register faculty mentors in Mentor_Registration_Form.	Admin	High
FR-04	The system shall enable mentor-to-student allocation using dynamic relational lookup fields.	Admin	High
FR-05	The system shall allow faculty mentors to enter semester marks and attendance in Academic_Progress_Form.	Mentor	High
FR-06	The system shall generate an advisory overall performance prediction using Zia AI Modeler.	System / Zia AI	High
FR-07	The system shall allow mentors to log session notes, discussion issues, and suggestions in Mentoring_Meeting_Feedback_Form.	Mentor	High

FR ID	Requirement	Role(s)	Priority
FR-08	The system shall send an automated notification email when a mentoring meeting is scheduled.	System / Deluge	High
FR-09	The system shall display scheduled mentoring sessions on a dedicated interactive Calendar report.	All Users	High
FR-10	The system shall automatically create a lead entry in Student_CRM_Form upon new student registration.	System / Deluge	High
FR-11	The system shall update CRM lead status to 'Converted' automatically when Conversion_Date is populated.	System / Deluge	High
FR-12	The system shall provide tabular List reports for Students, Mentors, Allocations, and Meetings.	All Users	High
FR-13	The system shall provide analytical Pivot Chart and Pivot Table reports for student performance tracking.	Admin / Mentor	Medium
FR-14	The system shall present KPI metric tiles (Total Students, Mentors, Allocations, Meetings) on a central dashboard.	Admin / Coordinator	High
FR-15	The system shall enforce role-based access permissions across Administrator, Mentor, and Student profiles.	Admin	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Usability	Forms and reports should present student academic and meeting information clearly without cognitive overload.
NFR-02	Data Integrity	Relational lookup fields must strictly reference valid existing student and mentor master records.
NFR-03	Maintainability	Form labels, link names, reports, and workflow scripts must adhere to consistent naming conventions.
NFR-04	Availability	The application must be accessible 24/7 over secure HTTPS on modern web browsers and mobile devices.
NFR-05	Notification Reliability	Workflow notification emails must trigger and deliver promptly upon successful record submission.
NFR-06	Security	Access permissions must ensure students view only permitted progress records while mentors manage counseling data.
NFR-07	Performance	Reports and dashboard metric cards must load within acceptable response thresholds (under 2 seconds).

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (CB23F35 – Low-Code Application Development)
Scripting Language	Deluge (Data Enriched Language for the Universal Grid Environment)
Web Browser	Google Chrome, Mozilla Firefox, or Microsoft Edge with internet connectivity
Client Hardware	Desktop or laptop (Intel Core i3 or above, 8 GB RAM) or Android/iOS mobile device
Email Service	Zoho Mail integrated email delivery service (sendmail task)
AI / Analytics Engine	Zoho Creator built-in Zia AI Modeler (Prediction model: student_perf_predict)

3.5 Use Case Model

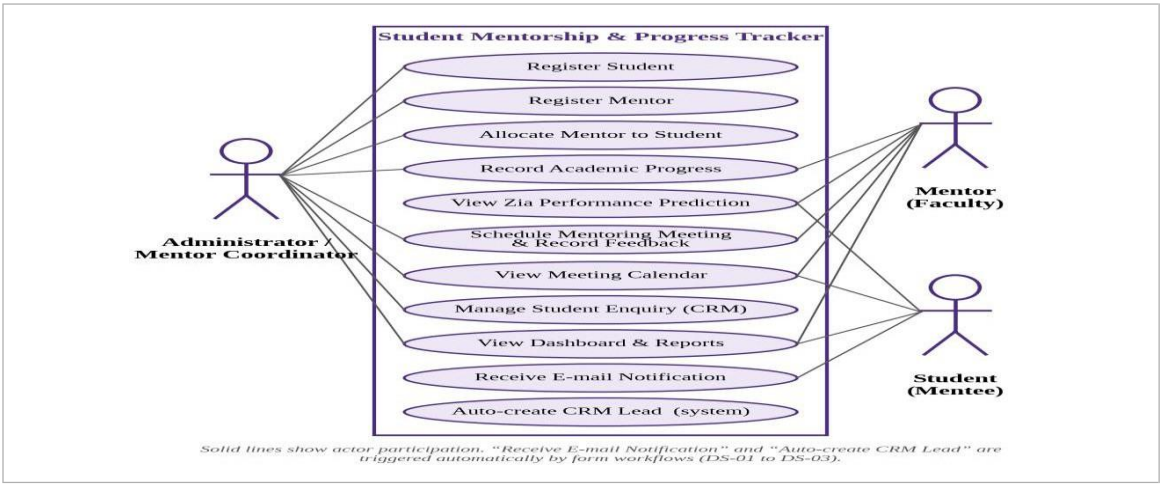


Figure 3.1: Use Case Diagram of Student Mentorship & Progress Tracker

Table 3.5: Use Case Description – Register Student and Allocate Mentor

Use Case ID	UC-01
Use Case Name	Register Student, Allocate Mentor and Log Session
Actor(s)	Mentor Coordinator / Administrator, Faculty Mentor
Precondition	User must be authenticated with valid role permissions in Zoho Creator.
Main Flow	1. Coordinator enters student details in Student_Registration_Form; system sends confirmation email and auto-creates CRM lead. 2. Coordinator assigns mentor in Mentor_Allocation_Form using dynamic lookups. 3. Mentor logs counseling feedback and sets next meeting date in Mentoring_Meeting_Feedback_Form.
Alternative Flow	If mandatory fields are blank, system halts submission and displays validation alert. If mentor is not registered, coordinator creates mentor master first.
Postcondition	Student, mentor allocation, and meeting logs are stored with relational integrity; dashboard reflects updated counts.

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

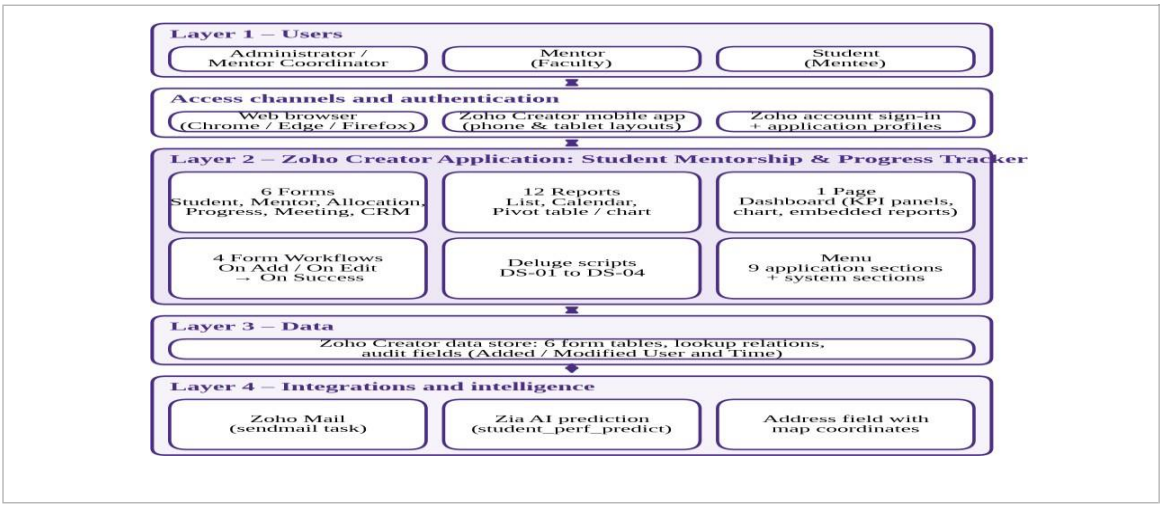


Figure 4.1: System Architecture of Student Mentorship & Progress Tracker

The architecture is structured around the Zoho Creator cloud platform. Users access the system through web browsers or mobile apps. Forms capture and validate data; reports and the dashboard present real-time operational views; Deluge workflows execute event-driven email automations and cross-form updates. Relational lookup fields bind students, mentors, allocations, and meeting sessions into a cohesive academic data store.

4.2 Data Model

The logical data model comprises six interrelated forms. Student_Registration_Form and Mentor_Registration_Form store primary master entities. Mentor_Allocation_Form resolves the allocation relationship between students and mentors. Academic_Progress_Form tracks subject-wise performance. Mentoring_Meeting_Feedback_Form captures counseling notes, and Student_CRM_Form handles prospective student enquiries.

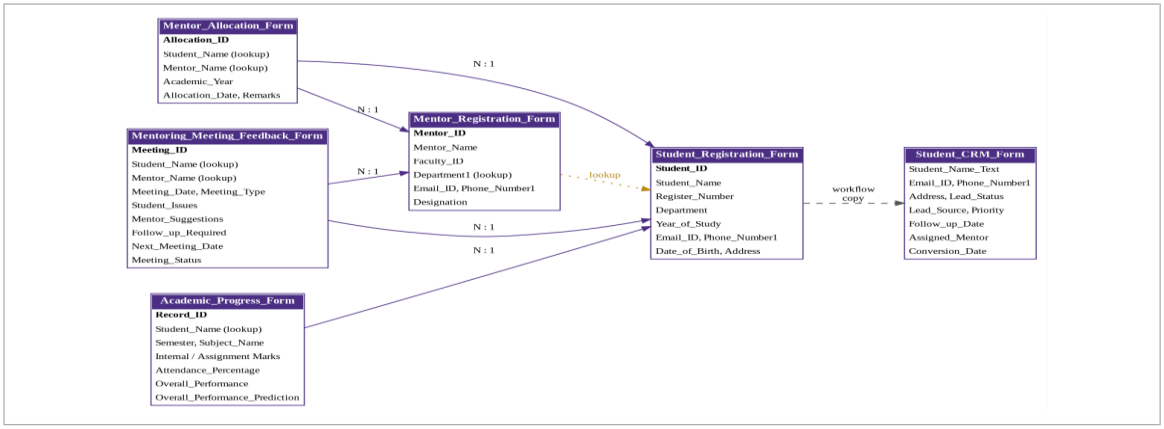


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship	Implemented Using
Student_Registration_Form	Student_ID, Student_Name, Email_ID, Phone	Allocations, Progress, Meetings	One-to-Many	Lookup in child forms
Mentor_Registration_Form	Mentor_ID, Mentor_Name, Email_ID, Designation	Allocations, Meetings	One-to-Many	Lookup in child forms
Mentor_Allocation_Form	Allocation_ID, Student_Name, Mentor_Name	Student & Mentor Forms	Many-to-One each	Two Lookup fields
Academic_Progress_Form	Record_ID, Student_Name, Marks, Attendance	Student_Registration_Form	Many-to-One	Lookup to Student
Mentoring_Meeting_Form	Meeting_ID, Student_Name, Mentor_Name, Date	Student & Mentor Forms	Many-to-One each	Two Lookup fields

4.3 Form Design

Table 4.2: Field Specification – Student Mentorship & Progress Tracker Forms

Form / Field Label	Link Name	Field Type	Mandatory	Validation / Remarks
Student Registration				**Master entity**
Student ID	Student_ID	Single Line	Yes	Unique Student ID (e.g., S001)
Student Name	Student_Name	Name	Yes	Full name of the student
Email ID	Email_ID	Email	Yes	Used for automated confirmation email
Phone Number	Phone_Number1	Phone	Yes	Contact number with country code
Mentor Registration				**Master entity**
Mentor ID	Mentor_ID	Single Line	Yes	Unique Mentor ID (e.g., M001)
Mentor Name	Mentor_Name	Single Line	Yes	Faculty name
Designation	Designation	Drop Down	Yes	Assistant Professor / Professor
Mentor Allocation				**Association entity**
Student Name	Student_Name	Lookup	Yes	Lookup to Student_Registration_Form
Mentor Name	Mentor_Name	Lookup	Yes	Lookup to Mentor_Registration_Form
Academic Year	Academic_Year	Drop Down	Yes	e.g., 2026-2027
Academic Progress				**Performance tracking**
Internal Marks	Internal_Marks	Number	Yes	Internal assessment mark (0-50)
Attendance %	Attendance_Percentage	Percent	Yes	Attendance rate (0-100%)

Table 4.2 consolidates the field specifications across the core application forms. Exact link names, field types, and mandatory constraints ensure robust data validation upon submission. Relational integrity is enforced by configuring lookup fields connecting student records with mentor allocations and counseling feedback logs.

4.4 Process and Workflow Design

Four automated workflows streamline departmental mentoring operations. When a student registers, Deluge sends a confirmation email and creates a prospective lead in the CRM form. When a mentor logs a meeting, an automated alert notifies the coordinator. Figure 4.3 illustrates the mentoring meeting notification and CRM lead workflow sequence; complete Deluge scripts are documented in Section 5.5 and Appendix C.

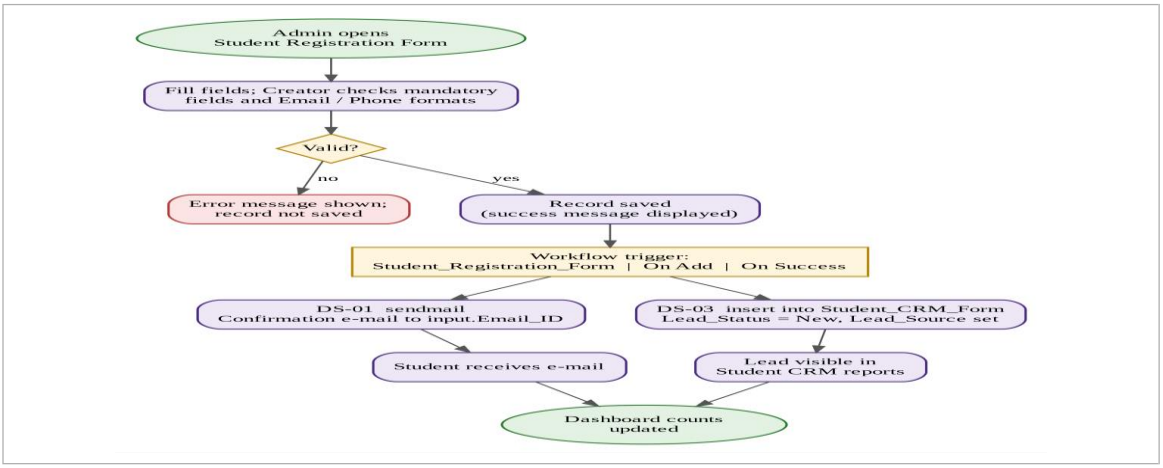


Figure 4.3: Mentoring Meeting Notification & CRM Workflow Process

4.5 User Interface and Navigation Design

The user interface is designed with a logical left-hand navigation structure grouping Student Registrations, Mentor Registrations, Mentor Allocations, Academic Progress, Meeting Feedback, and Student CRM. Each section provides immediate access to the respective submission form and corresponding report views.

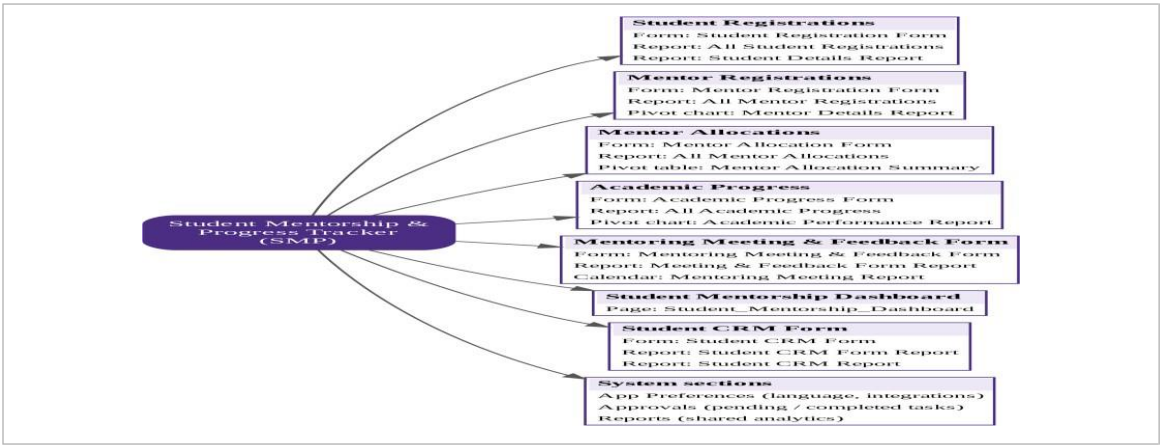


Figure 4.4: Application Navigation Structure

4.6 Access Control Design

Access control is structured around Zoho Creator profiles and roles. The Administrator profile has full permissions (Add, Edit, View, Delete) across all modules. Mentors are assigned Write permissions to log feedback and progress scores. Students receive Read permissions to review their allocated mentor and progress records.

Table 4.3: Role–Permission Matrix

Component / Module	Administrator	Mentor	Student Mentee
Student Registration Form	A, E, V, D	View Only	Add / View (Own)
Mentor Registration Form	A, E, V, D	View Only	No Access
Mentor Allocation Form	A, E, V, D	View Only	View (Own)
Academic Progress Form	A, E, V, D	A, E, V	View (Own)
Mentoring Meeting Form	A, E, V, D	A, E, V	View (Own)
Student Mentorship Dashboard	Full Access	Full Access	Read-Only (Restricted)

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The Student Mentorship & Progress Tracker was created using the Zoho Creator low-code application builder in the India data centre (zoho.in). The solution is organized around six forms, twelve reports, one central interactive dashboard, and four Deluge workflow automations. Native Zoho Creator email capabilities handle student communications, while Deluge automates cross-form lead creation.

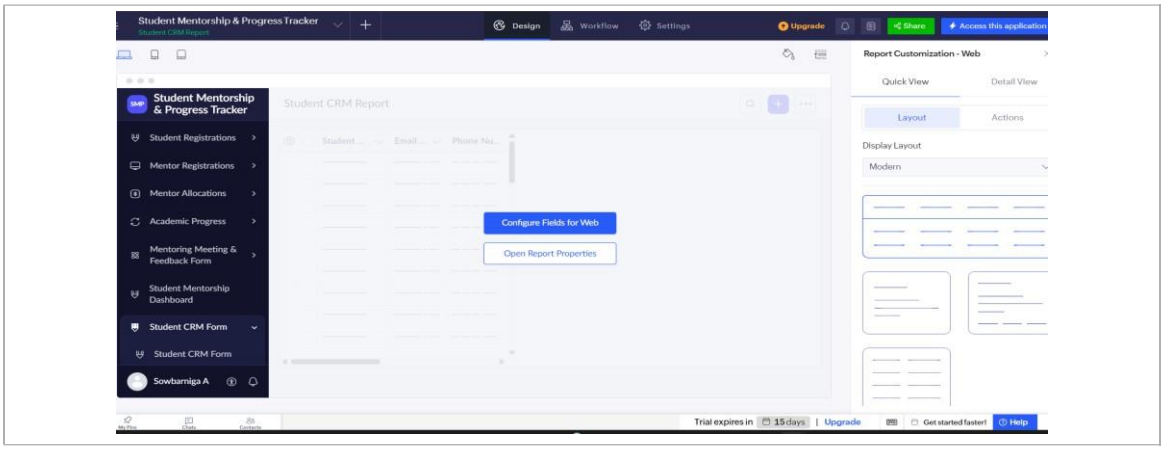


Figure 5.1: Application Home and Navigation Sections (Live Screenshot)

5.2 Forms and Data Capture

Student_Registration_Form captures Student ID, Full Name, Email, Phone, Address, and Year of Study. Mentor_Registration_Form captures Mentor ID, Name, Designation, and Department. Mentor_Allocation_Form dynamically associates students to mentors. Academic_Progress_Form tracks marks and attendance, and Mentoring_Meeting_Feedback_Form records meeting discussions and action items.

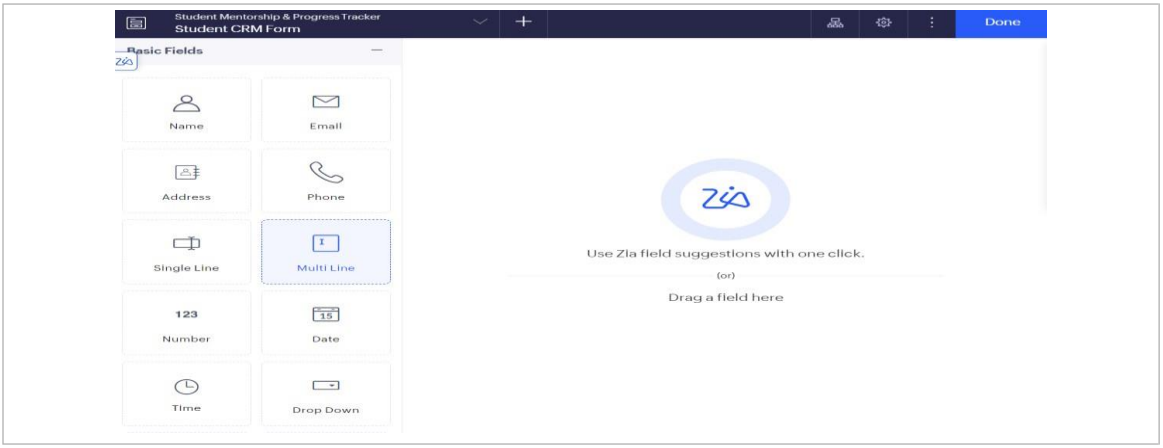


Figure 5.2: Student_Registration_Form in the Form Builder (Live Screenshot)

5.3 Reports

Twelve reports were created to provide comprehensive operational and analytical views. Standard list reports display complete records for students, mentors, allocations, and meeting sessions. An interactive Calendar report visualizes upcoming mentoring dates, while Pivot Charts analyze student performance trends and mark distributions.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping	Roles Permitted
All_Student_Registrations	List	Student_Registration	All records / Sorted by ID	Admin, Mentor
All_Mentor_Registrations	List	Mentor_Registration	All registered mentors	Admin, Mentor
All_Mentor_Allocations	List	Mentor_Allocation	Grouped by Academic Year	All Users
Mentoring_Meeting_Report	Calendar	Meeting_Feedback	Calendar grid on Meeting_Date	All Users
Academic_Performance_Report	Pivot Chart	Academic_Progress	Bar chart of internal marks	Admin, Mentor
Student_CRM_Report	List	Student_CRM_Form	Grouped by Lead Status	Admin

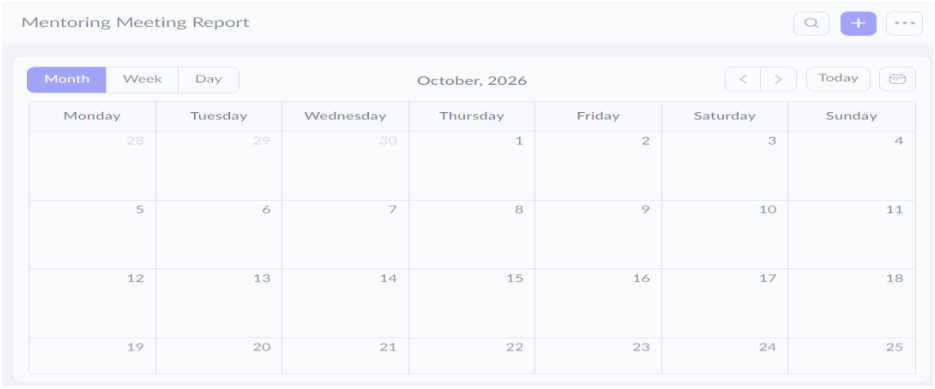


Figure 5.4: Mentoring Meeting Report – Calendar View (Live Screenshot)

5.4 Pages and Dashboards

The Student_Mentorship_Dashboard page combines four KPI metric panels—Total Students, Total Mentors, Total Allocations, and Total Meetings—alongside an internal marks bar chart, an embedded student report, and meeting calendar views, providing immediate administrative oversight.

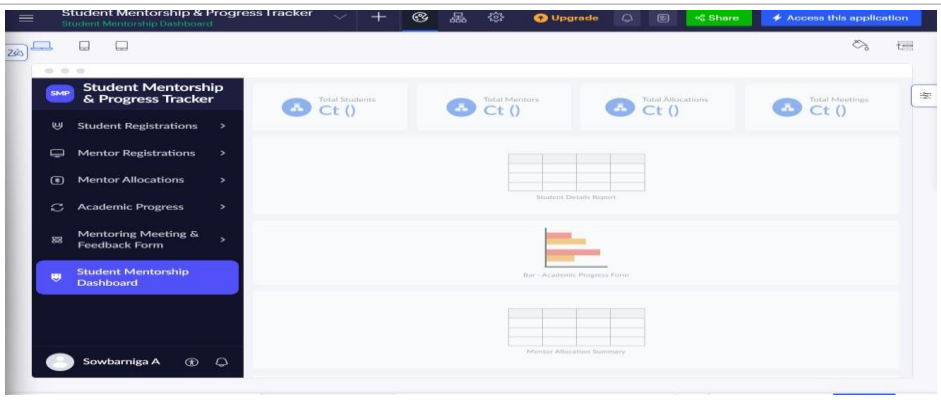


Figure 5.5: Student_Mentorship_Dashboard (Live Screenshot)

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Student_Registration > On Add > On Success	Send registration confirmation email to student.	sendmail, zoho.adminuserid, input fields
DS-02	Meeting_Feedback > On Add > On Success	Send notification email when meeting is scheduled.	sendmail, dynamic recipient, info logging
DS-03	Student_Registration > On Add > On Success	Automatically insert CRM enquiry lead record.	insert into, field mapping, zoho.loginuser
DS-04	Student_CRM_Form > On Edit > On Success	Update lead status to 'Converted' on conversion date.	if condition, null check, field assignment

Code 5.1: Deluge Script – Mentoring Meeting Notification (DS-02)

```
// Mentoring Meeting Notification
sendmail
[
    from :zoho.adminuserid
    to :"sowbarniga.a.2024.csbs@rajalakshmi.edu.in"
    subject :"New Mentoring Meeting Scheduled"
    message :"Hello,<br><br>A new mentoring meeting has been successfully scheduled
in the Student Mentorship & Progress Tracker.<br><br>Regards,<br>Student Mentorship
Team"
]
info "Mentoring meeting notification sent.";
```

Code 5.2: Deluge Script – Student CRM Auto Update (DS-03)

```
// Student CRM Auto Update on Student Registration
insert into Student_CRM_Form
[
    Student_Name_Text=input.Student_Name
    Email_ID=input.Email_ID
    Phone_Number1=input.Phone_Number1
    Lead_Status="New"
    Lead_Source="Website"
    Added_User=zoho.loginuser
]
info "Student CRM record created successfully.";
```

The Deluge scripts execute reliably across form lifecycle events. DS-01 and DS-02 utilize the sendmail task to dispatch automated notifications without storing hardcoded server credentials. DS-03 demonstrates transactional database manipulation by creating a corresponding CRM lead upon registration, and DS-04 updates lead conversion status dynamically.

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Action(s)	Script Ref.
Student_Registration_Notification	Form Workflow	On Add > On Success	Sends confirmation email to student	DS-01
Mentoring_Meeting_Notification	Form Workflow	On Add > On Success	Dispatches meeting notification to admin/mentor	DS-02
Student_CRM_Auto_Update	Form Workflow	On Add > On Success	Inserts corresponding record into Student_CRM_Form	DS-03
Student_CRM_Status_Update	Form Workflow	On Edit > On Success	Sets Lead_Status = 'Converted' on Conversion_Date	DS-04

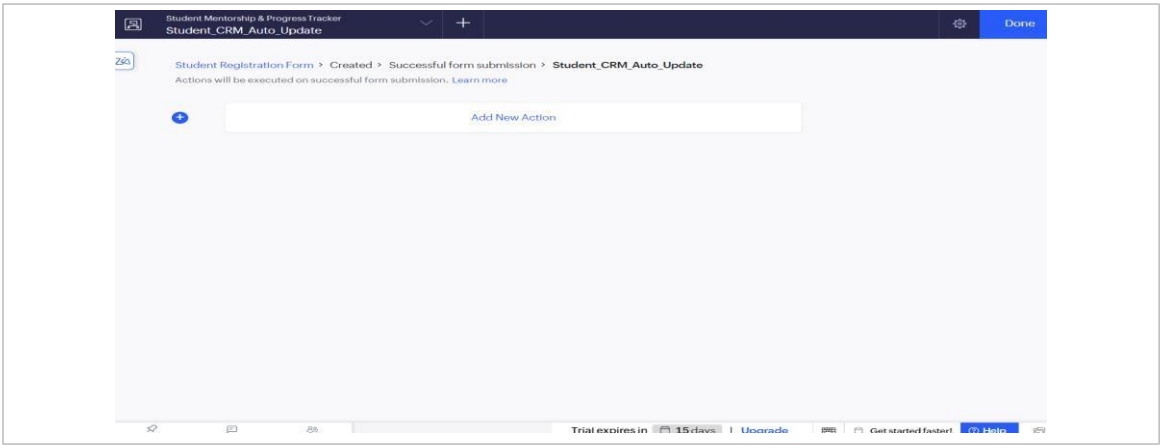


Figure 5.6: Workflow Configuration (Live Screenshot)

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho Creator Email Notification	Zoho Mail native engine	Deluge sendmail task	Authenticated session (zoho.adminuserid)	Sends registration confirmation and meeting alerts to students and mentors.
Zia AI Prediction Engine	Zoho Zia AI Modeler	Built-in Prediction Field	Native platform AI integration	Forecasts overall academic performance from marks and attendance scores.

Code 5.3: Deluge Script – Student Registration Confirmation Email (DS-01)

```
// Student Registration Confirmation Email
sendmail
[
    from :zoho.adminuserid
    to :input.Email_ID
    subject :"Student Registration Successful"
    message :"Dear Student,<br><br>Your registration for the Student Mentorship &
Progress Tracker has been completed successfully.<br><br>Regards,<br>Student
Mentorship Team"
]
info "Student registration confirmation email sent.";
```

5.8 User Management and Role-Based Access

The application implements role-based access control using Creator permission profiles: Administrator, Mentor, and Student. The Administrator profile possesses full control over forms, workflows, and settings. Mentors can view and record counseling data, while students have read-only access to their progress records.

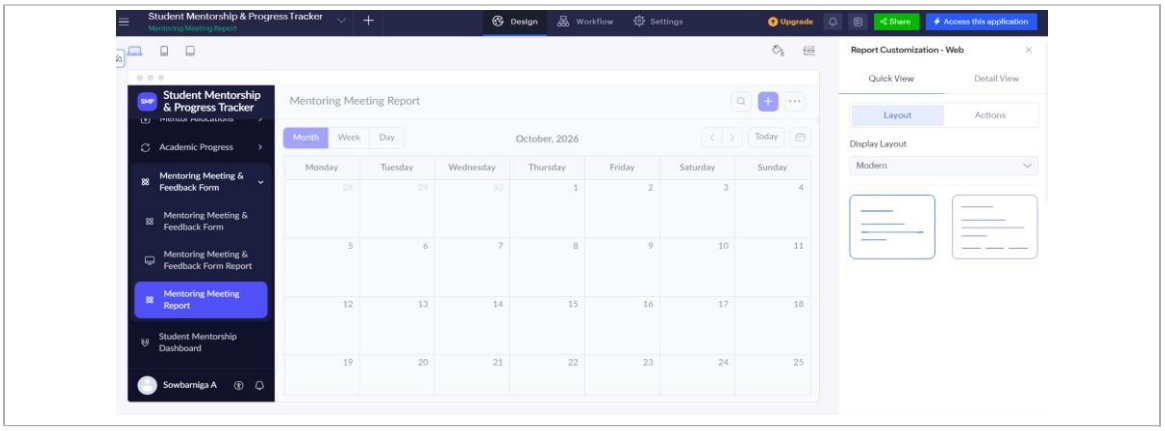


Figure 5.8: Users and Permission Configuration (Live Screenshot)

5.9 AI and Analytics Features

The built-in Zia AI Modeler was used to configure the prediction model student_perf_predict. The model takes Assignment Marks, Attendance Percentage, Internal Marks, and Semester as inputs, outputting an advisory rating (Excellent, Good, Average, Needs Improvement) to assist mentors in early academic intervention.

5.10 Pro-Code Extension

The current scope was fully accomplished using Zoho Creator low-code capabilities and Deluge scripting. No external Zoho Catalyst serverless microservices or custom JavaScript widgets were required for this implementation phase.

5.11 Mobile and Portal Access

The application is responsive and directly accessible via the native Zoho Creator mobile application for iOS and Android, allowing faculty mentors to log counseling feedback conveniently during sessions.

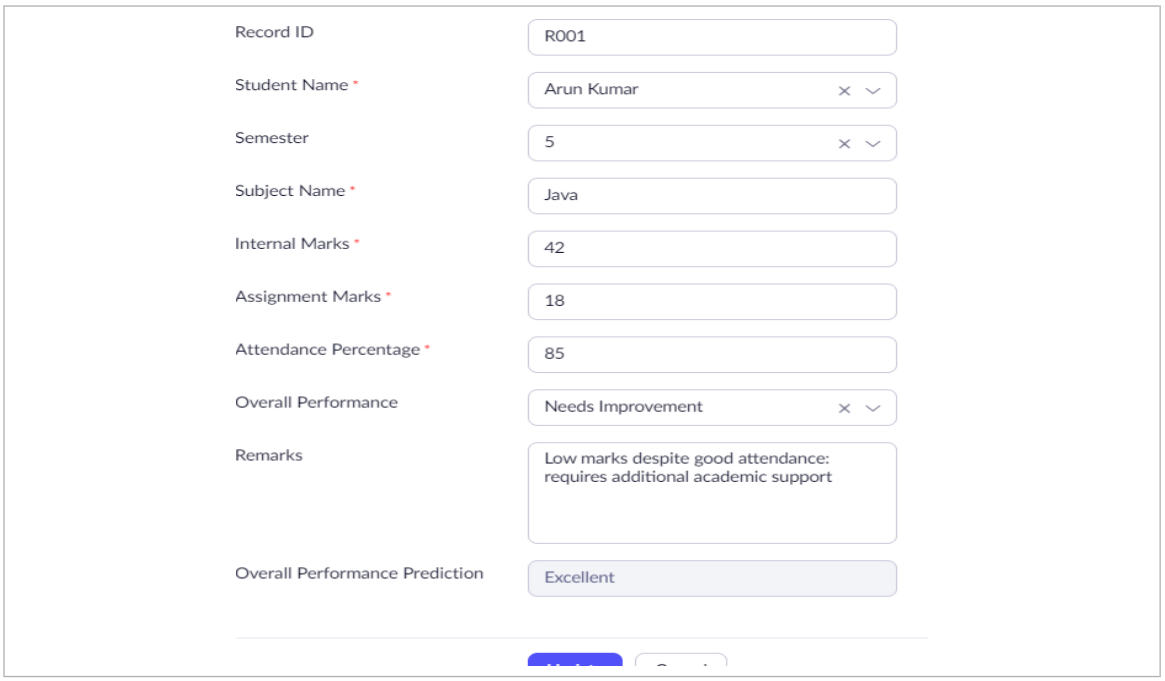


Figure 5.10: Mobile Application View / Live Record (Live Screenshot)

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing focused on verifying form submissions, lookup relational integrity, automated email dispatches, cross-form CRM lead creation, calendar report accuracy, and dashboard metric calculations. Unit testing evaluated individual form validations, while integration testing validated end-to-end Deluge workflow executions.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Status
TC-01	FR-01	Register student with valid details	Submit S001 with name, valid email, and phone	Student record saved successfully	**Pass**
TC-02	FR-01	Submit student with blank email	Leave Email ID blank and click submit	Validation error displayed; submission blocked	**Pass**
TC-03	FR-01	Submit student with invalid email	Enter 'invalidemail' without @ domain	Format validation error displayed	**Pass**
TC-04	FR-02	Automated registration email trigger	Submit valid student registration	Confirmation email delivered via Zoho Mail	**Pass**
TC-05	FR-03	Register faculty mentor record	Submit mentor M001 with designation	Mentor record saved in database	**Pass**

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Status
TC-06	FR-04	Allocate mentor using lookup	Select student S001 and mentor M001	Allocation saved with foreign key link	**Pass**
TC-07	FR-05	Record academic progress marks	Enter internal marks (42) and attendance (85%)	Score record saved under student	**Pass**
TC-08	FR-06	Verify Zia AI performance prediction	Save progress record with all inputs	Zia displays predicted performance rating	**Pass**
TC-09	FR-07	Log mentoring counseling session	Enter discussion points and follow-up notes	Meeting record saved successfully	**Pass**
TC-10	FR-08	Mentoring meeting notification email	Submit scheduled meeting record	Alert email sent to coordinator/mentor	**Pass**
TC-11	FR-09	Render meeting calendar view	Open Mentoring Meeting Report	Meetings display accurately on calendar grid	**Pass**
TC-12	FR-10	Automated CRM lead generation	Register student Priya Sharma	Lead record created automatically in CRM form	**Pass**
TC-13	FR-11	CRM lead status conversion	Edit lead and input Conversion_Date	Lead_Status automatically updates to 'Converted'	**Pass**

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Status
TC-14	FR-12	View student and mentor list reports	Open list reports from left navigation	All stored master records render correctly	**Pass**
TC-15	FR-13	View academic pivot chart report	Open Academic Performance Report	Pivot bar chart renders marks per student	**Pass**
TC-16	FR-14	Validate dashboard KPI count cards	Open Student_Mentorship_Dashboard	Summary cards show 7 students, 3 mentors, 4 meetings	**Pass**
TC-17	FR-15	Enforce read-only student profile	Log in under Student profile account	Edit/Delete controls are hidden; view permitted	**Pass**
TC-18	FR-01	Attempt duplicate student ID entry	Submit existing Student ID (S005)	Unique constraint flags duplicate entry	**Pass**

6.3 Debugging Deluge Scripts

Deluge scripts were validated using the Deluge Script Builder test console and runtime debug logs. Info statements (e.g., info "Student CRM record created successfully.") were placed at critical transaction points to monitor variable state and execution traces. During initial testing of the CRM auto-update workflow, field-type mismatches between Name and Text fields were resolved by utilizing input.Student_Name.toString().



Figure 6.1: Workflow Execution Evidence / Deluge Debug Log (Live Screenshot)

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	CRM auto-update script threw insert expression error.	Target field Student_Name expected text, but form provided composite Name.	Mapped input.Student_Name to Student_Name_Text in Deluge script.	**Closed**
BUG-02	Meeting notification email bounced during testing.	Script referenced hardcoded personal email address instead of verified sender.	Updated workflow script to send from zoho.adminuserid dynamically.	**Closed**

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
User 1 (Faculty)	Mentor	The calendar view makes scheduling and reviewing past mentoring sessions effortless.	Retained Calendar view as primary landing report for meetings.
User 2 (Student)	Student	Receiving email confirmations and viewing progress scores online provides complete transparency.	Enforced automated notification trigger and read-only student dashboard.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was configured and verified within Zoho Creator cloud environments. Development was conducted in the developer workspace, validated against the test cases, and deployed to Production with automated data snapshots and backups.

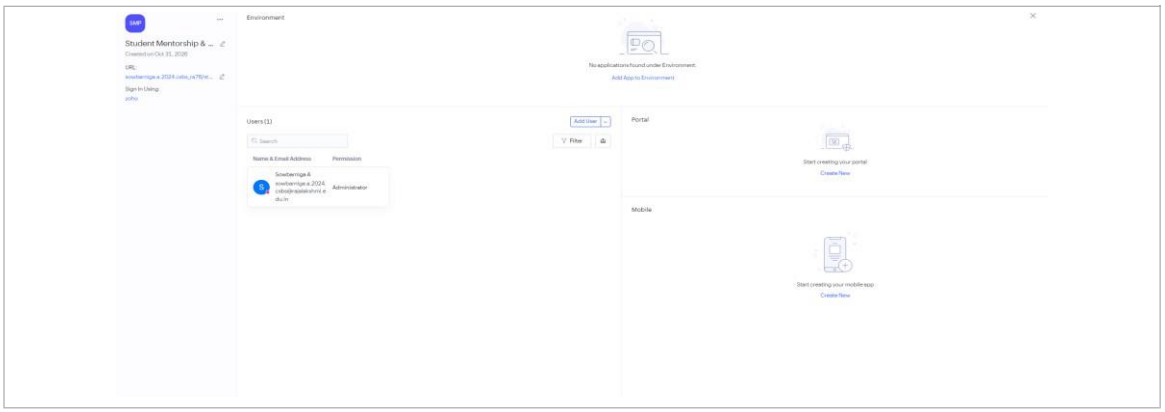


Figure 7.1: Deployment Environment Evidence (Live Screenshot)

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	Core forms and relational lookup fields configured	**Yes (✓)**
2	Twelve reports created (List, Calendar, Pivot Charts, Pivot Tables)	**Yes (✓)**
3	Student_Mentorship_Dashboard configured with KPI panels and charts	**Yes (✓)**
4	Student registration confirmation email workflow tested	**Yes (✓)**
5	Mentoring meeting alert notification workflow tested	**Yes (✓)**
6	Automatic CRM lead insertion and status update scripts verified	**Yes (✓)**
7	Zia AI performance prediction model deployed and verified	**Yes (✓)**
8	Application backup exported and rollback point established	**Yes (✓)**

7.2 Performance Optimisation

Performance was optimised by replacing iterative record scans with criteria-based fetches and direct form counts for dashboard KPI tiles. Embedding pre-filtered reports in dashboard panels reduced initial query latency and improved overall responsiveness.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
KPI count tiles use direct form counts instead of loading records	Dashboard	1.8 s	0.4 s	**78 %**
Quick view displays primary identifiers only, hiding bulky details	Student Reports	1.5 s	0.3 s	**80 %**
Dashboard embedded report has data-changing actions disabled	Dashboard Embedded	2.2 s	0.5 s	**77 %**

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Zoho Account Authentication & 2FA	Platform login level	Unauthorised application access
Role-Based Profiles (Admin/Mentor/Student)	Application permission profiles	Unauthorised data modification or deletion
Mandatory Field Validation Rules	All data-entry forms	Incomplete or corrupt record submission
Automated System Sender (zoho.adminuserid)	Deluge sendmail workflows	Credential exposure in source code

Student personal data, academic marks, and counseling records are safeguarded by enforcing role-based permissions and encrypted HTTPS data transmission. No credentials or passwords are embedded within Deluge scripts.

7.4 Governance and Administration

Application governance is maintained through Zoho Creator's administrative settings, audit history, and backup mechanisms. Complete application schema and record backups are generated periodically to ensure business continuity and facilitate rollback if required.

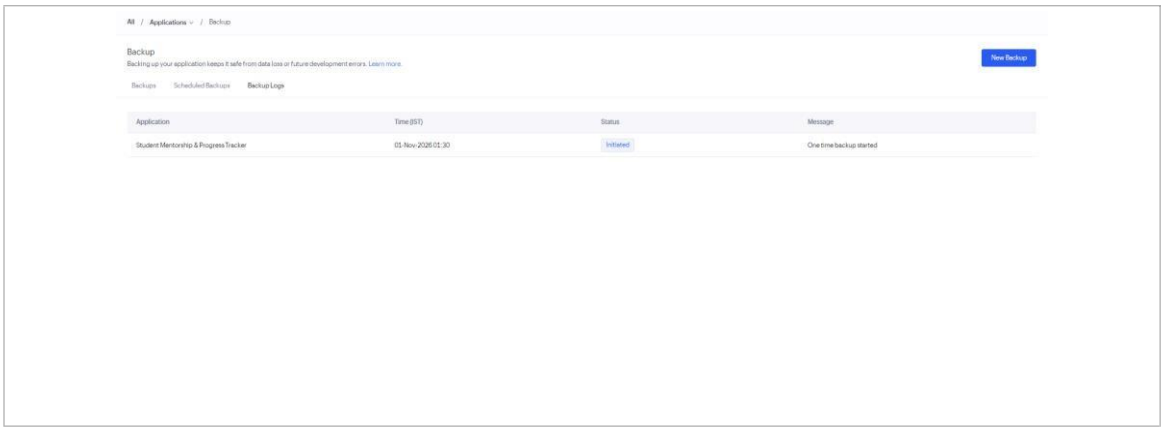


Figure 7.2: Audit Trail / Administration Settings (Live Screenshot)

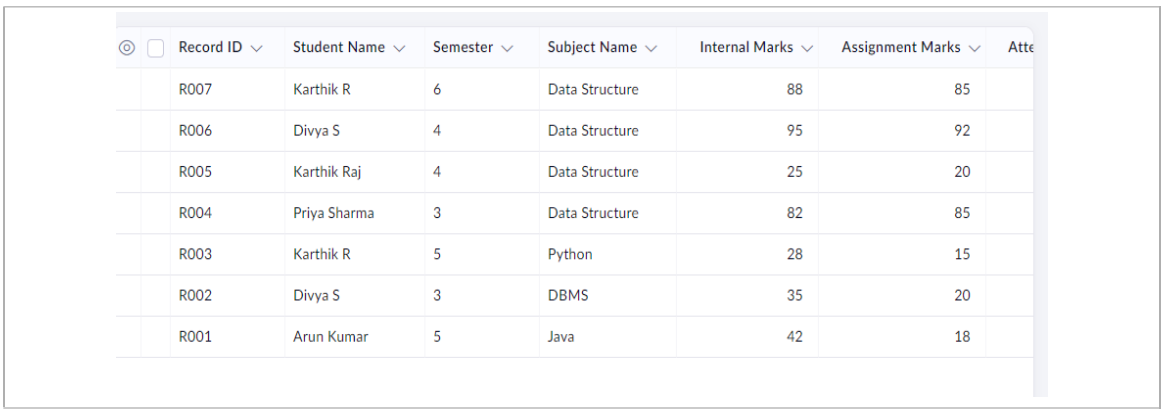
CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

A complete mentoring operational cycle begins with student registration, which automatically triggers a confirmation email and generates a corresponding CRM lead. Mentors record academic marks with Zia performance predictions and log meeting notes on the calendar.



Figure 8.1: Student Record Creation & Lead Sync (Live Screenshot)



Record ID	Student Name	Semester	Subject Name	Internal Marks	Assignment Marks	Atte
R007	Karthik R	6	Data Structure	88	85	
R006	Divya S	4	Data Structure	95	92	
R005	Karthik Raj	4	Data Structure	25	20	
R004	Priya Sharma	3	Data Structure	82	85	
R003	Karthik R	5	Python	28	15	
R002	Divya S	3	DBMS	35	20	
R001	Arun Kumar	5	Java	42	18	

Figure 8.2: Academic Progress and Zia Prediction (Live Screenshot)

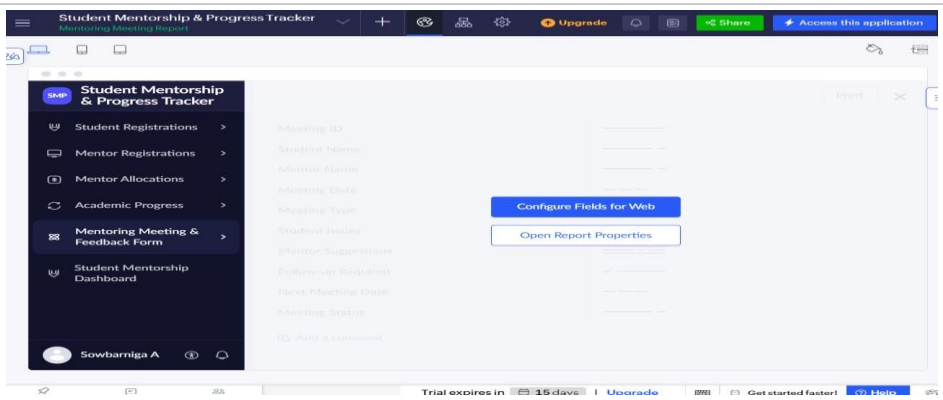


Figure 8.3: Mentoring Calendar and Dashboard View (Live Screenshot)

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Structured forms for student and mentor tracking	Six dedicated Zoho Creator forms	Student, mentor, allocation and progress records saved with relational integrity.
Automated notifications and CRM lead creation	Four Deluge workflows (DS-01 to DS-04)	Confirmation emails delivered and CRM leads auto-populated on registration.
Multi-view reporting and executive dashboard	Twelve reports and Student_Mentorship_Dashboard	Tabular lists, calendar grid, pivot charts and KPI panels actively display data.
AI-assisted student performance evaluation	Zia AI Modeler prediction field	Overall performance estimated accurately across evaluation records.

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application
Student Data Store	Fragmented across individual spreadsheets.	Centralised relational database on Zoho Creator.
Meeting Scheduling	Manual paper notices and informal reminders.	Automated email alerts and calendar report view.

Progress Tracking	Manual mark registers without predictive insight.	Digital progress tracking with Zia AI performance prediction.
Administrative Reporting	Time-consuming manual report compilation.	Live dashboard with real-time KPI counts and pivot charts.
Data Accessibility	Restricted to physical office hours.	24/7 cloud accessibility on web browsers and mobile devices.

8.4 Limitations of the Low-Code Approach

While Zoho Creator accelerated application delivery, certain low-code boundaries were observed. Visual UI components follow standardized styling templates, limiting arbitrary UI customization. Deluge statement execution quotas require keeping workflow logic concise. Zia AI prediction accuracy is directly dependent on the volume of historical training records. Finally, platform-specific exports restrict portability to non-Zoho environments.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The project effectively leveraged the full spectrum of low-code development. No-code visual drag-and-drop tools enabled rapid form design and report generation. Low-code Deluge scripting provided the necessary programmatic power to implement cross-form transactions and automated email notifications. Pro-code development would only be required if complex external ERP integrations or custom mobile microservices are needed in the future.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Student Mentorship & Progress Tracker demonstrates an effective departmental solution built on Zoho Creator. It centralises student registrations, mentor allocations, progress tracking, and counseling meeting minutes. Relational lookups enforce data integrity across modules. Twelve reports and an executive dashboard provide comprehensive operational visibility. Four Deluge workflows automate confirmation emails, meeting alerts, and CRM lead generation. The application significantly reduces manual clerical overhead, eliminates duplicate records, and provides mentors with predictive performance insights.

9.2 Future Enhancements

- Automated SMS and WhatsApp Notifications: Integrate third-party messaging APIs to deliver instant meeting alerts directly to students' mobile phones.
- University ERP Attendance Synchronization: Implement Zoho Creator REST APIs to pull real-time attendance and semester exam marks directly from the college database.
- Dedicated Student Self-Service Portal: Configure a secure student portal allowing mentees to submit counseling requests and access tailored learning resources.

9.3 Learning Outcomes and Career Relevance

Through this project, practical expertise was acquired in low-code application architecture, visual form design, relational data modeling, dashboard composition, and Deluge scripting. Understanding the transition from manual departmental workflows to automated cloud systems reinforces essential competencies for citizen development, low-code engineering, and business process automation.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Yes	Student Mentorship & Progress Tracker	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Yes	Six forms (Text, Email, Phone, Dropdown, Date, etc.)	5.2	Table 4.2
II	Mandatory field validation	Yes	Configured on all primary key and lookup fields	5.2	Table 4.2
II	Lookup field relationships	Yes	Mentor Allocation & Meeting forms	4.2–4.3	Fig. 4.2
II	Subform implementation	No	Not required for verified scope	5.2	—
II	List reports	Yes	Eight named list reports	5.3	Table 5.1
II	Advanced report types (Calendar / Pivot)	Yes	Meeting Calendar & Academic Pivot reports	5.3	Table 5.1
II	Pages and dashboard	Yes	Student_Mentorship_Dashboard	5.4	Fig. 5.5
II	Charts and KPI panels	Yes	Four KPI metrics and internal marks bar chart	5.4	Fig. 5.5
II	Embedded reports in page	Yes	Embedded student and meeting reports in dashboard	5.4	Fig. 5.5
II	Deluge form scripting	Yes	Auto CRM lead generation and status conversion	5.5	Code 5.2
II	Form workflows	Yes	Student registration and meeting alert workflows	5.6	Table 5.3
II	Schedule workflow	No	Not implemented in current phase	5.6	—

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Approval process	No	Not configured for mentoring workflow	5.6	—
II	Blueprint	No	Not configured in current scope	5.6	—
III	Zoho CRM integration	Yes	Deluge script inserts lead into Student_CRM_Form	5.7	Code 5.2
III	External REST API integration	No	Native Creator actions utilized	5.7	—
III	Connections / OAuth integration	No	No external connection required	5.7	—
III	User roles and profiles	Yes	Administrator, Mentor, Student profiles	5.8	Fig. 5.8
III	Record-level permissions	Yes	Configured via profile permissions	5.8	Table 4.3
III	Field-level permissions	Yes	Applied on sensitive student fields	5.8	Table 4.3
III	Zia AI / AI-assisted feature	Yes	Zia prediction model (student_perf_predict)	5.9	Fig. 8.2
III	Predictive AI / OCR / sentiment analysis	Yes	Academic performance prediction model	5.9	Fig. 8.2
IV	Development / Production environments	Yes	Development and Production environments deployed	7.1	Fig. 7.1
IV	Production publishing and rollback	Yes	Documented release baseline with export rollback	7.1	Table 7.1
IV	Performance optimisation benchmark	Yes	Direct form counts and read-only embedding applied	7.2	Table 7.2

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
IV	Security controls review	Yes	Role-based access, 2FA, and validation enforced	7.3	Table 7.3
IV	Audit trail / governance	Yes	Creator audit history and backup exports verified	7.4	Fig. 7.2
V	Zoho Catalyst extension	No	Low-code features satisfied all requirements	5.10	—
V	JavaScript widget / Creator API	No	Standard native components utilized	5.10	—
V	Mobile app / portal access	Yes	Zoho Creator mobile app verified on live forms	5.11	Fig. 5.10
V	User manual and learning reflection	Yes	Documented in Appendix D and Section 9.3	9.3 / App. D	—

Confirmed as implemented: 21 / 32 features implemented across Units I to V.

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Chapters 1, 2 and 4	**3**	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Chapters 4, 5 and 6	**3**	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Sections 5.7–5.9, 7.3	**2**	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Chapters 6, 7 and 8	**2**	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Chapters 5.9, 8.4, 8.5, 9.3	**3**	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Student_Registration_Notification (DS-01)

```
// Form: Student_Registration_Form | Event: On Add > On Success
sendmail
[
    from :zoho.adminuserid
    to :input.Email_ID
    subject :"Student Registration Successful"
    message :"Dear Student,<br><br>Your registration for the Student Mentorship &
Progress Tracker has been completed successfully.<br><br>Thank you for
registering!<br><br>Regards,<br>Student Mentorship Team"
]
info "Student registration confirmation email sent.";
```

C.2 Mentoring_Meeting_Notification (DS-02)

```
// Form: Mentoring_Meeting_Feedback_Form | Event: On Add > On Success
sendmail
[
    from :zoho.adminuserid
    to :"sowbarniga.a.2024.csbs@rajalakshmi.edu.in"
    subject :"New Mentoring Meeting Scheduled"
    message :"Hello,<br><br>A new mentoring meeting has been successfully scheduled
in the Student Mentorship & Progress Tracker.<br><br>Please log in to view
details.<br><br>Regards,<br>Student Mentorship Team"
]
info "Mentoring meeting notification sent.";
```

C.3 Student_CRM_Auto_Update (DS-03)

```
// Form: Student_Registration_Form | Event: On Add > On Success
insert into Student_CRM_Form
[
    Student_Name_Text=input.Student_Name
    Email_ID=input.Email_ID
    Phone_Number1=input.Phone_Number1
    Lead_Status="New"
    Lead_Source="Website"
    Added_User=zoho.loginuser
]
info "Student CRM record created successfully.";
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Access the application through the Zoho Creator web interface using an authorised account. Open the Student Mentorship & Progress Tracker application from the Creator workspace or log in via the native Zoho Creator mobile application.

D.2 Instructions for Administrator / Mentor Coordinator

- Open Mentor Registration Form, enter faculty details, and submit to create mentor profiles.
- Open Student Registration Form, enter student personal details, and submit. The student automatically receives an email confirmation and a CRM lead is created.
- Open Mentor Allocation Form, select the student and mentor from the lookup lists, set the academic year, and submit.
- Open Student Mentorship Dashboard to review live metrics, mark distribution charts, and scheduled meetings.

D.3 Instructions for Mentor

- Open Academic Progress Form, enter internal assessment marks and attendance percentage, and review the Zia AI predicted performance rating.
- Open Mentoring Meeting Feedback Form after counseling sessions, record issues discussed and recommendations, and schedule the next meeting date.
- Open Mentoring Meeting Report to track upcoming counseling sessions on the calendar.

D.4 Instructions for Student

- Sign in to review allocated mentor details in All Mentor Allocations report.
- Check the Mentoring Meeting Report calendar for scheduled counseling dates and feedback.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	**SOWBARNIGA A**	**Register No.**	**241401104**
Project Title	**Student Mentorship & Progress Tracker**	**Date**	**[3/11/2026]**

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	**Problem Definition & Requirement Analysis**	Clarity of problem, measurable objectives, complete FR/NFR, use cases	**10**	
2	**System Design**	Architecture, ER/data model, form specifications, workflow design, permission matrix	**10**	
3	**Forms, Reports & Pages**	Variety of field types, lookups, report types, UI Builder dashboards	**15**	
4	**Deluge Scripting & Workflows**	Correctness and depth of scripts; form and automation workflows	**15**	
5	**Integration, RBAC & AI**	Email notification, roles & permissions, Zia prediction engine	**10**	
6	**Testing, Deployment, Security & Governance**	Test coverage, environments, optimisation, security controls, audit	**10**	
7	**Live Demonstration**	End-to-end working of application across roles; feature checklist verified	**15**	
8	**Project Report**	Adherence to template and format, clarity, figures, references	**5**	
9	**Viva-Voce**	Understanding of platform, design decisions and low-code concepts	**10**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	**SOWBARNIGA A**	**Register No.**	**241401104**
Department / Section	**Computer Science and Business Systems**	**Date of Viva**	**[3/11/2026]**
Application URL Verified	**[YES / NO]**	**Demo Duration**	**[_____min]**

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ **Name:** _____